

AYUSH TRIVEDI

AI Solution Architect | Agentic AI & RAG Systems | GenAI Engineering
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EXECUTIVE SUMMARY

Claude Certified Architect (Anthropic, CCA-F) — Senior AI Consultant and Solution Architect with 5+ years designing and shipping production-grade agentic AI and RAG platforms for regulated enterprises. Core specialization: **multi-agent orchestration via the Model Context Protocol (MCP) and pydantic-ai**, paired with hand-architected retrieval pipelines on AWS (Bedrock, SageMaker, S3, Lambda) rather than off-the-shelf managed services — delivering USD 4.4M in validated annual operational savings at Amazon and governed agentic platforms at Deloitte USI across pharmaceutical and insurance verticals. Deep expertise in responsible AI governance: human-in-the-loop checkpoints, prompt-level safety guardrails, structured-output validation, and audit-trail design for zero-tolerance compliance environments. Combines boardroom-level strategic framing with the engineering depth to own delivery end to end.

CERTIFICATION

- **Claude Certified Architect – Foundations (CCA-F), Anthropic** | 2026 | View Credential — enterprise agentic architecture, MCP orchestration, RAG system design, context engineering, AI safety & governance, responsible AI evaluation, observability, and human-in-the-loop system design.

PROFESSIONAL EXPERIENCE

Deloitte USI | Senior AI Consultant & Solution Architect

Mar 2026 – Present | Bengaluru

Leading enterprise AI advisory and solution architecture for global B2B clients — translating ambiguous transformation goals into governed, production-ready agentic platforms across regulated industries.

- Defined enterprise AI strategy and end-to-end architecture for a flagship agentic intelligence platform on Amazon EKS — a **hub-and-spoke multi-agent topology**: a pydantic-ai orchestrator core coordinating a dedicated MCP server ecosystem (Alation, Database, Datastore, GraphQL MCP) alongside a Databricks Bronze/Silver/Gold medallion pipeline with an AI Agent Enrichment layer (Glossary, Graph, Lineage, Profiling agents) — turning a fragmented data landscape into a governed, queryable knowledge fabric.
- Advised on responsible AI governance and trust architecture: structured human-in-the-loop review checkpoints, prompt guardrails, and explainability frameworks meeting enterprise auditability and compliance standards prior to production release.
- Engineered a semantic knowledge layer combining **pg-vector (dense retrieval) and Neo4j Knowledge Graph (GraphRAG)** for ontology-aware metric resolution and enterprise-wide data lineage — enabling natural-language queries over cross-domain operational data with citation-backed, auditable responses.
- Championed AI adoption and executive alignment — led discovery workshops, translated board-level transformation objectives into phased delivery roadmaps, and communicated architectural trade-offs to technical and non-technical leadership alike.

Client Engagements (Confidential)

Global Pharmaceutical Enterprise — AI Strategy & Regulatory Automation

Pharma & Life Sciences | Agentic AI | RAG | Responsible AI | Regulatory Compliance

- **Challenge:** mounting pressure to accelerate regulatory submission timelines while managing vast, unstructured clinical and drug data across siloed systems — with zero tolerance for non-compliant or hallucinated outputs in a heavily audited environment.
- **Approach:** co-architected an LLM-integrated agentic pipeline with MCP-based tool orchestration for automated drug data processing and regulatory document generation. Designed RAG pipelines over proprietary pharmaceutical knowledge bases with multi-layer responsible AI guardrails — prompt-level safety constraints, structured output validation, and mandatory human-in-the-loop review gates before any regulatory content reached end-users.
- **Outcome:** reduced regulatory document preparation effort by an estimated 40–50%, accelerating submission readiness. Established an AI governance framework — data sensitivity classifications, audit trail requirements, model explainability standards — that earned sign-off from the client's compliance and legal leadership.

Leading North American Insurance Carrier — Intelligent Underwriting & Claims AI

Insurance & Financial Services | Agentic AI | Document Intelligence | AI Governance

- **Challenge:** high underwriting cycle times and manual bottlenecks in claims processing — limiting throughput, increasing cost, and creating inconsistent decision quality across geographies.
- **Approach:** designed and delivered an agentic architecture combining MCP multi-agent orchestration for enterprise system integration, LLM-driven document understanding for extraction and classification of policy/claims documents, and structured output validation with full audit trails. Embedded confidence scoring and human-review flagging for low-certainty decisions, aligned with insurance-sector data privacy regulations.
- **Outcome:** projected 30–35% reduction in underwriting cycle time and a significant decrease in manual claims triage effort, positioning the client for phased autonomous decision-making with human oversight retained for high-value and complex claims.

Architected and deployed production-grade GenAI and RAG automation systems on AWS — translating audit complexity into governed, scalable workflows delivering USD 4.4M in annual savings.

- **Architected a hand-built, hybrid-retrieval RAG pipeline** for audit automation — ingesting structured/unstructured documents via Amazon S3, embedding via SageMaker-hosted HuggingFace sentence transformers, indexing into FAISS and Pinecone vector stores, and orchestrating retrieval-augmented generation through the **Bedrock Converse/InvokeModel API** invoked from AWS Lambda — delivering citation-backed, auditable audit decisions and cutting manual review effort by 70%+.
- Designed a multi-stage document intelligence pipeline using Amazon Textract for structured extraction from scanned audit files, S3-triggered Lambda preprocessing, and Bedrock-hosted LLM inference for classification and decision support — enabling automated NWR/RCR audit validation at scale and a 94% workflow efficiency improvement.
- Drove USD 4.4M in validated annual operational savings by replacing manual audit lookup and review workflows with a governed RAG knowledge base on AWS (S3 + SageMaker + Pinecone + Bedrock Claude); reduced capital expenditure by 15% through a data-driven, ML-informed budget allocation model.

PowerSchool | Associate Application Engineer

Dec 2021 – Apr 2023 | Bengaluru

- Developed modular ML data pipelines (Python/SQL) for feature engineering, model training support, and analytics integration within a B2B SaaS platform; contributed to data quality validation and EDA workflows for downstream model evaluation.

HCL Tech | Analyst Trainee

Feb 2021 – Dec 2021 | Bengaluru

- Built and validated large-scale structured datasets using Python and SQL to support analytics reporting and performance monitoring pipelines.

KEY ACHIEVEMENTS & BUSINESS IMPACT

- **USD 4.4M Annual Savings (Validated):** delivered AWS-native RAG audit automation (S3 + SageMaker + Pinecone + Bedrock) eliminating high-volume manual review — Amazon's most impactful AI automation program in the audit vertical.
- **94% Audit Workflow Automation:** multi-stage document intelligence pipeline (Textract + Lambda + Bedrock) achieving near-full autonomous NWR/RCR audit validation at enterprise scale.
- **Pharma Regulatory Acceleration:** AI advisory engagement estimated to reduce regulatory submission preparation effort by 40–50%, with a client-approved governance framework covering auditability, safety guardrails, and compliance sign-off.
- **Insurance Underwriting Efficiency:** agentic architecture projected to reduce underwriting cycle time by 30–35% and substantially decrease manual claims triage across geographies.
- **Enterprise Agentic Platform Delivery:** architected a full-stack agentic intelligence platform on Amazon EKS — pydantic-ai agent core, MCP server ecosystem, Databricks medallion data platform, Neo4j knowledge graph, and AWS Bedrock LLM integration.

CORE SKILLS & TECHNICAL EXPERTISE**Agentic Orchestration & Multi-Agent Systems — Production Experience**

pydantic-ai • MCP (Model Context Protocol) • Multi-Agent Orchestration (hub-and-spoke, pipeline patterns) • Agent Design Patterns • Human-in-the-Loop Systems • Context Engineering • Prompt Architecture • AI Safety Guardrails • Explainability & Auditability • LLM Observability

RAG, Knowledge Graphs & Semantic Search — Production Experience

End-to-End RAG Pipeline Design • Hybrid Retrieval • HuggingFace Sentence Transformers • FAISS • Pinecone • pg-vector • Neo4j Knowledge Graph (GraphRAG) • Unstructured Data Ingestion • Amazon Textract • Tesseract OCR • Knowledge Base Design

Cloud AI & Data Platforms — Production Experience

AWS Bedrock (Converse / InvokeModel API) • AWS SageMaker (hosted embedding models) • Amazon EKS • AWS Lambda • Amazon S3 • Databricks (Unity Catalog, Medallion Architecture) • SQLGlue • Alation Data Catalog • Snowflake • Oracle • PostgreSQL • Redis

Working Knowledge / Secondary Frameworks

LangChain • LangSmith • OpenAI SDK — evaluated and used for specific tooling and tracing tasks; primary orchestration work is done via pydantic-ai and MCP above.

AI Strategy & Enterprise Advisory

AI Transformation Roadmaps • Executive Stakeholder Alignment • Business-to-Technical Translation • Discovery Workshops • AI Adoption Strategy • ROI Framing • AI Governance Frameworks • Responsible AI • Risk-Aware Architecture

Engineering & Platform Delivery

Python • FastAPI • REST API Orchestration • Docker • Logfire Observability • Pydantic • SQL • Git • CI/CD for AI • TensorFlow • Scikit-learn

Programme Management & Regulated Industries

Agile Delivery • Technical Roadmapping • Backlog Prioritization • Release Management • Risk & Dependency Management • Executive Reporting • B2B SaaS • Pharmaceutical AI • Insurance Technology • Regulatory-Aware AI Systems • Compliance-Driven Architecture

EDUCATION

B.Tech – Electronics and Communication Engineering | PSIT Kanpur | 2016 – 2020